Topology optimized bowsprit for an 18-footer

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The situation

An 18-footer skiff carries a large gennaker on a forward **bowsprit**. Rope tension and sail aerodynamic forces demand a spar that is stiff and light.

Figure 1: 18-footer skiff with gennaker. Bowsprit projects forward; sail tack attaches at tip. Photo: MySailing [1].

Problem definition

Modelled as a **2D plane-stress** rectangular cantilevered beam made from aluminum with the properties $E_0 = 70$ GPa, $\nu = 0.33$, thickness $t = 0.05$ m and minimum volume fraction $V_f = 0.25$. The domain is $3.80m \times 0.50m \ \ (L \times H)$ with the forces: $\boldsymbol{F}_1 = 2$ kN at 65° on a small strip $0.175L$ to $0.225L$ and $\boldsymbol{F}_2 = 3$ kN at 55° on a small strip $0.95L$ to L as seen in Figure 2.

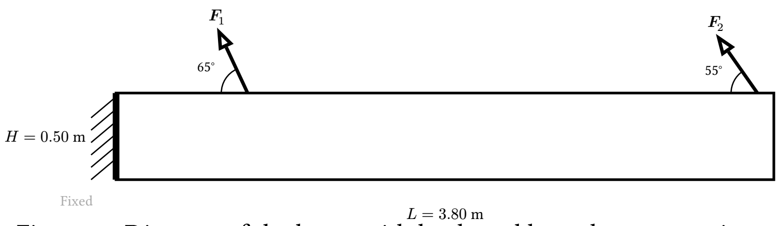

Figure 2: Diagram of the beam with loads and boundary constraints.

Elasticity BVP - weak form

Find $\boldsymbol{u} \in V_0$ such that $\forall \boldsymbol{v} \in V_0$:

$$\int_\Omega \boldsymbol{\sigma}(\boldsymbol{u}) : \boldsymbol{\varepsilon}(\boldsymbol{v})\,\mathrm{d}\Omega = \int_{\Gamma_N} \boldsymbol{T} \cdot \boldsymbol{v}\,\mathrm{d}\Gamma$$

Small-strain, plane-stress: $\boldsymbol{\sigma} = \lambda_{\text{eff}}\ \mathrm{tr}(\boldsymbol{\varepsilon})\boldsymbol{I} + 2\mu\boldsymbol{\varepsilon}$, $\lambda_{\text{eff}} = \frac{E\nu}{1-\nu^2}$. BC: $\boldsymbol{u} = \boldsymbol{0}$ on Γ_D; tractions $\boldsymbol{T}$ on Γ_N.

Compliance (cost function) = external work: $J = \int_{\Gamma_N} \boldsymbol{T} \cdot \boldsymbol{u}\,\mathrm{d}\Gamma$

Optimization problem

Design variable $\rho \in [0, 1]$ element-wise (DG0).

$$\min_{\rho} \begin{cases} J_d + J_i + J_e & \text{Task 3} \\ J_d + J_e & \text{Task 4} \end{cases} \quad \text{s.t.} \quad \begin{cases} \frac{1}{LH}\int_\Omega \bar{\rho}_d\,\mathrm{d}\Omega \le V_f \\ \frac{\int_\Omega w\bar{\rho}_d\,\mathrm{d}\Omega}{\int_\Omega w\,\mathrm{d}\Omega} \le V_f \\ \sigma_c^k \le 1 \quad \forall k \\ 0 \le \rho_e \le 1 \end{cases}$$

Pitch weight $w = 1 + 2.5\left(\frac{x}{L}\right)^2$. Constraints on **dilated** $\bar{\rho}_d$ guarantee all realizations satisfy limits. Gradients via pyadjoint; solved by MMA.

SIMP - stiffness interpolation

$$E_e = E_{\min} + \bar{\rho}_e^p(E_0 - E_{\min}), \quad p = 3, \quad E_{\min} = 10^{-6}E_0$$

$p = 3$ penalises intermediate densities; $E_{\min}$ prevents a singular stiffness matrix in voids.

Double-filter robust formulation

Three realizations from a single ρ via a two-stage pipeline:

$$\rho \xrightarrow{\text{filter}} \tilde{\rho}_1 \xrightarrow{\eta_{\min}} \bar{\rho}_1 \xrightarrow{\text{filter}} \tilde{\rho}_2 \xrightarrow{\eta_k} \bar{\rho}_k$$

Filters solved in CG1 (radius $r = 0.04$ m); projections in DG0.

Helmholtz PDE filter (weak form)

$$\int_\Omega r^2 \nabla\tilde{\rho} \cdot \nabla v\,\mathrm{d}\Omega + \int_\Omega \tilde{\rho}, v\,\mathrm{d}\Omega = \int_\Omega \rho, v\,\mathrm{d}\Omega \quad \forall v$$

with $\partial\tilde{\rho}/\partial n = 0$ on $\partial\Omega$. Enforces minimum feature size $\approx 2r = 0.08$ m. CG1 trial space required (DG0 gradients vanish).

Heaviside projection

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\rho} - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}$$

Sharpness β ramped $(1, 2, \ldots, 432)$, 15 stages, 30 MMA iters/stage: $\eta_d = 0.45$ (**dilated**), $\eta_i = 0.50$ (**nominal**), $\eta_e = 0.55$ (**eroded**); $\Delta\eta = 0.05$. Shared load path = **manufacturing-robust** design.

Stress constraint - P-mean aggregation

Yield stress $\sigma_y = 40$ MPa. One constraint per realization:

$$\sigma_c^p = \frac{1}{|\Omega|}\int_\Omega \left(\frac{\sigma_m}{\alpha\sigma_y}\right)^p \mathrm{d}\Omega \le 1$$

ε-relaxation removes void singularity ($\varepsilon = 0.2$):

$$f_\sigma = \frac{\bar{\rho}}{\varepsilon(1 - \bar{\rho}) + \bar{\rho}}, \quad \sigma_m = f_\sigma\sqrt{\sigma_{\text{vM}}^2 + \sigma_{\min}^2}$$

$$\sigma_{\text{vM}}^2 = \sigma_{11}^2 + \sigma_{22}^2 - \sigma_{11}\sigma_{22} + 3\sigma_{12}^2$$

β cap $\beta \le 2R/l_e \approx 9.6$ prevents artificial stress concentrations. **p-continuation**: p ramped $2 \to 400$ over 15 stages.

Optimized designs

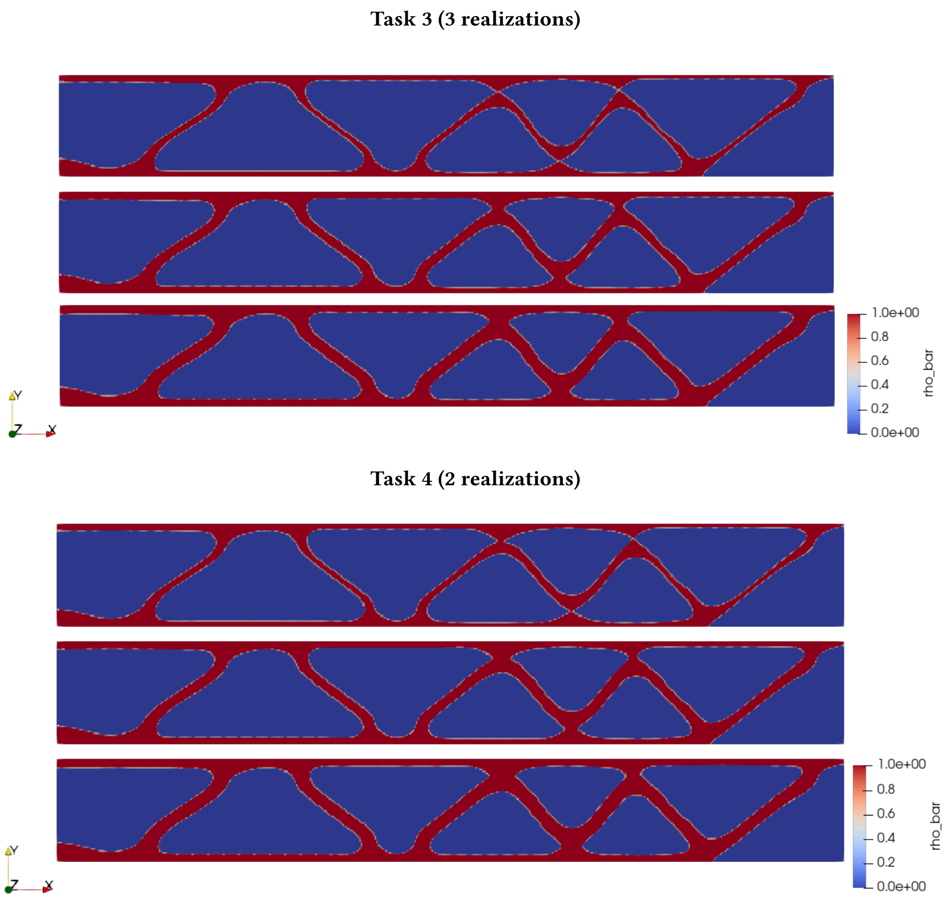

Figure 3: Optimized density fields. Top to bottom per task: eroded ($\eta_e = 0.55$), nominal ($\eta_i = 0.50$, Task 3 only), dilated ($\eta_d = 0.45$). Blue = solid, white = void.

Results & conclusion (Task 5)

Design	J (N·m)	V	V_p	σ_c
T3 dilated	10.09	.299†	.291†	0.619
T3 nominal	11.65	.257†	.248†	0.664
T3 eroded	14.82	.204	.193	**1.126**†
T4 dilated	10.51	.301†	.295†	0.654
T4 interm.	12.03	.260†	.253†	0.729
T4 eroded	15.41	.207	.199	**1.152**†

†constraint violated ($V_f = 0.25$, $\sigma_c \le 1$). Task 3 nominal J is 3.3% lower than Task 4 intermediate - including the nominal realization yields a stiffer design. SIMP, Helmholtz filtering, and double Heaviside projection produce a lightweight bowsprit; stress constraints remain binding in the eroded realization.

[1] MySailing, "18ft Skiffs Spring Championship Race 2." 2024.